

Cactus health: rot, spots & stress

Read the pattern before naming a pathogen. Similar symptoms can have very different causes.

First decide what is changing

A firm old scar, a fresh sunburn and rapidly collapsing tissue require different responses. Photograph the whole plant and the affected area with a scale, note the date, and compare the boundary over time. Isolate unexplained progressive damage before sharing tools or cuttings.

Soft or collapsing

Act promptly: isolate, keep runoff contained, assess roots and stem. Rapid wet decay warrants professional diagnostic advice, particularly if several plants are affected.

Firm, dry and stable

Record and monitor. Do not cut out every spot; unnecessary wounds create another problem.

New distortion or mottling

Check pests and environmental changes. Hold off on propagation while an unexplained pattern is assessed.

Old basal corking

Gradual, firm basal tissue can be normal aging. An expanding soft margin or sudden decline changes the assessment.

UC IPM · Cactus pests and disorders · UC IPM · Houseplant problems

Four different causes

Physiological injury

Sun, frost, water imbalance, salts or physical damage can injure tissue without a primary infection. Record recent changes rather than diagnosing from color.

True fungi

Different fungi cause spots, cankers and rots. A fungus name requires supporting identification; “black spot” is a symptom description.

Bacteria

Soft-rot bacteria can cause rapidly macerating tissue. Nutrition may influence susceptibility, but calcium supplementation is not proof of a cure.

Oomycetes

Phytophthora and *Pythium* are water moulds, distinct from true fungi. Treatment selection must match the organism and label; one generic fungicide is not a universal solution.

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Read the host on the paper

Grafted Gymnocalycium A Korean study identified *Pectobacterium brasiliense* in soft rot of grafted cactus. A confirmed host-pathogen report is stronger evidence than a diagnosis based on an internet photo.

Dragon fruit A Malaysian study identified *Neoscytalidium dimidiatum* and reproduced canker symptoms in red-fleshed dragon fruit. It does not show that every orange or black spot on a *Trichocereus* has that cause.

Park et al. · *Pectobacterium* on grafted *Gymnocalycium* (2022) · Mohd et al. · Dragon-fruit stem canker (2013)

Viruses: appearance cannot clear a plant

A 2018 investigation detected six virus species in a single *Notocactus leninghausii* f. *cristatus*. This demonstrates possible mixed infection, not its prevalence in all cultivated cacti. A visually clean cutting is not a virus test, and a mottled plant is not automatically infected.

For unexplained systemic symptoms, pause exchanges and grafting, use dedicated tools and consult a plant diagnostic laboratory about suitable sampling and tests. Surface cleaning and removing a damaged tip cannot be assumed to eliminate a systemic infection.

Park et al. · Six viruses detected in one cactus (2018)

When salvage is reasonable

For a localized rot problem, an experienced grower may attempt to retain clearly sound tissue. Separate the plant first; use clean equipment, remove contaminated material and avoid carrying sap between plants. If discoloration continues through the stem or the cause is unclear, seek diagnosis before propagating it.

For ordinary stem cuttings, allow the wound to dry and callus before rooting. A fresh graft is different: its cut surfaces need immediate contact. Neither cinnamon nor a casual sulfur dip is a validated substitute for tool sanitation. Fresh media and clean containers reduce carryover risk.

RHS · Growing cacti and succulents

Document the growing conditions

Log root condition, recent rain or irrigation, pot drying time, temperature, light changes and any sprays or feed. This is the useful part of a whole-plant approach: cultural stress and infection can interact. It does not make all soft rot a calcium deficiency or justify adding fertilizer to an unexplained collapsing plant.

Continue to GeeBee: practice and evidence or the pest inspection guide.

Interactive inspection assistant and current source links: revolutionarydesigns.io/cactus/guides/fungal.html